

ENCH296 2026 Exam - Question 5

Sucrose (table sugar, $C_{12}H_{22}O_{11}$) is the most common dietary sugar. When metabolised by the body it is ultimately oxidised to CO_2 and H_2O — a process energetically equivalent to combustion. Food scientists measure the energy content of foods by burning samples in a bomb calorimeter: the food is ignited inside a sealed steel vessel (the “bomb”) filled with oxygen, which is itself immersed in a water bath. The entire assembly sits inside a perfectly insulated outer container. After the reaction is complete, the temperature rise of the water bath is used to determine the enthalpy of combustion.

The following data are provided:

Species	Enthalpy of formation, ΔH_f° / kJ mol ⁻¹	Heat capacity, C_P / J mol ⁻¹ K ⁻¹	Molecular weight / g mol ⁻¹
$C_{12}H_{22}O_{11}$ (s)	-2222	425.5	342
O (g)	0	29.4	32
CO ₂ (g)	-393.5	37.1	44
H ₂ O (g)	-241.8	33.6	18
H ₂ O (l)	-285.8	75.4	18

Component	Mass / kg	Heat capacity (C_P) / kJ kg ⁻¹ K ⁻¹
Steel bomb	0.5	0.49
Water bath	2.0	4.18

The bomb is charged with 5× the stoichiometric quantity of oxygen. Mass of sucrose sample: 20 g. The system starts at 298.15 K.

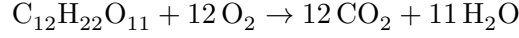
- Write the balanced combustion reaction for sucrose.
- Calculate the enthalpy of reaction (ΔH_{rxn}°) at 298.15 K with $H_2O(g)$ as the product.

(c) Starting from the general energy balance, write down and simplify the energy balance that describes what happens inside the bomb calorimeter from ignition until the system reaches its final temperature.

(d) Calculate the final temperature of the entire system (bomb and water bath).

Solution

(a) Balanced combustion reaction



(b) Enthalpy of reaction at 298.15 K, with H₂O(g) as product

$$\Delta H_{rxn}^\circ = \sum \nu_j \Delta H_{f,j}^\circ = 12(-393.5) + 11(-241.8) - (-2222) - 0$$

$$\Delta H_{rxn}^\circ = -4722 - 2659.8 + 2222 = -5159.8 \text{ kJ/mol}$$

$$\Delta H_{rxn}^\circ \approx -5160 \text{ kJ/mol} \quad (\text{strongly exothermic})$$

Note: This is the value per mole of sucrose reacted.

(c) Simplified energy balance

Assumptions and simplifications:

- Perfectly insulated outer container $\Rightarrow \dot{Q} = 0$.
- Sealed vessel, no flow in or out $\Rightarrow$ all $\dot{m}H$ terms are zero.
- No work is done $\Rightarrow \dot{W} = 0$.
- Integrating the balance from ignition to final equilibrium $\Rightarrow \Delta H_{sys} = 0$ over the process.

Starting from the general energy balance:

$$\frac{d(m_{sys}H_{sys})}{dt} + \sum \dot{m}_{out}H_{out} - \sum \dot{m}_{in}H_{in} = \dot{Q} + \dot{W}$$

with the flow terms and $\dot{Q}$, $\dot{W}$ all zero, so:

$$\frac{d(m_{sys}H_{sys})}{dt} = 0$$

$$d(m_{sys}H_{sys}) = 0 \, dt$$

$$\int_i^f d(m_{sys}H_{sys}) = 0$$

$$m_{sys} \int_i^f dH_{sys} = 0$$

$$m_{sys} \Delta H_{sys} = 0$$

$$\Delta H_{sys} = 0$$

Following a reaction-then-heat path (react at 298.15 K, then bring everything to T_f), this becomes:

$$\Delta H_{sys} = n_{sucrose} \Delta H_{rxn,298.15} + \sum n_j C_{P,j} (T_f - T_i) + \sum m_k C_{P,k} (T_f - T_i) = 0$$

where the first sum runs over the gas-phase species in the bomb (combustion products plus excess O_2), and the second sum runs over the steel bomb and the water bath.

(d) Final temperature To calculate the final temperature, solve:

$$\begin{aligned} n_{sucrose} \Delta H_{rxn,298.15} + \sum n_j C_{P,j} (T_f - T_i) + \sum m_k C_{P,k} (T_f - T_i) &= 0 \\ \sum n_j C_{P,j} (T_f - T_i) + \sum m_k C_{P,k} (T_f - T_i) &= -n_{sucrose} \Delta H_{rxn,298.15} \\ (T_f - 298.15) (\sum n_j C_{P,j} + \sum m_k C_{P,k}) &= -n_{sucrose} \Delta H_{rxn,298.15} \\ T_f - T_i &= \frac{-n_{sucrose} \Delta H_{rxn,298.15}}{\sum n_j C_{P,j} + \sum m_k C_{P,k}} \\ T_f &= \frac{-n_{sucrose} \Delta H_{rxn,298.15}}{\sum n_j C_{P,j} + \sum m_k C_{P,k}} + T_i \end{aligned}$$

Moles of sucrose:

$$n_{sucrose} = \frac{20}{342} = 0.05848 \text{ mol}$$

Stoichiometric O_2 required: $12 \times 0.05848 = 0.7018$ mol. Total O_2 charged ($5 \times$ stoichiometric): 3.509 mol. Excess O_2 : $3.509 - 0.7018 = 2.807$ mol.

Products formed: $CO_2 = 0.7018$ mol; $H_2O(g) = 11 \times 0.05848 = 0.6433$ mol.

“Heat” released by the reaction:

$$n_{sucrose} \times \Delta H_{rxn}^\circ = 0.05848 \times -5159.8 = -301.745 \text{ kJ}$$

Heat capacity contributions of everything that must be warmed from 298.15 K to T_f :

Component	n or m	C_P	nC_P or mC_P (kJ/K)
CO ₂ (g)	0.7018 mol	0.0371 kJ mol ⁻¹ K ⁻¹	0.026
H ₂ O (g)	0.6433 mol	0.0336 kJ mol ⁻¹ K ⁻¹	0.022
Excess O ₂ (g)	2.807 mol	0.0294 kJ mol ⁻¹ K ⁻¹	0.082
Steel bomb	0.5 kg	0.49 kJ kg ⁻¹ K ⁻¹	0.245
Water bath	2.0 kg	4.18 kJ kg ⁻¹ K ⁻¹	8.360
Total			8.735

The gas-phase heat capacity contribution (0.130 kJ/K) is only about 1.5% of the total, so it is reasonable to neglect it, simplifying the total to $\sum m_k C_{P,k} = 8.605$ kJ/K (steel bomb + water bath only). Using the full total of 8.735 kJ/K:

$$\Delta T = \frac{-n_{\text{surcose}} \Delta H_{\text{rxn}, 298.15}}{\sum (n_j C_{P,j} + m_k C_{P,k})} = \frac{301.745}{8.735} = 34.54 \text{ K}$$

$$T_f = 298.15 + 34.54 = 332.69 \text{ K}$$

$$T_f = 332.69 \text{ K (59.5 } ^\circ\text{C)}$$